

Newly rendered players always look south until they move their head

MC-109346 Networking

Status	Resolved / Fixed
Affected	Minecraft 1.10.2, Minecraft 16w42a, Minecraft 1.12.1, Minecraft 1.12.2, Minecraft 18w21b, Minecraft 18w22a (+70 more)
Fixed in	1.20.2 Pre-release 2
Reported	2016-10-28

STEPS TO REPRODUCE

1. Connect Player A to a multiplayer server.
2. Connect Player B to the same server.
3. Have Player A face north and remain stationary.
4. Have Player B move far enough away that Player A is beyond render distance.
5. Have Player B return until Player A is rendered again.
6. Observe that Player B sees Player A facing south, even though Player A is still facing north and has not moved their head.

OBSERVED BEHAVIOUR

When a remote player is newly spawned or rendered on the client - such as after joining the server, entering render distance, or respawning - the client displays the player's body at the correct position and body rotation, but initializes the player's head rotation to south. As a result, the player's body can face one direction while their head and apparent looking direction face south.

This incorrect head rotation persists until the remote player changes their view or otherwise sends a head-rotation update. The issue is visible only to clients that newly render the player; the affected player's own view is not necessarily incorrect.

EXPECTED BEHAVIOUR

When a player is newly rendered, their body and head should immediately appear facing the direction they are actually looking, rather than defaulting to south. This orientation should remain accurate when the player joins the server, enters another player's render distance, or respawns, and should not require the rendered player to move their head before the correct direction is displayed.

ENVIRONMENT

Minecraft Java Edition Versions: 1.10.2, 16w42a, 1.12.1, 1.12.2, 18w21b, 18w22a, 18w22b, 18w22c, 1.13-pre1, 1.13-pre2, 1.13-pre4, 1.13-pre5, 1.13-pre7, 1.13, 18w30b, 1.13.1, 1.13.2-pre1, 1.13.2, 18w45a, 18w46a, 19w08b, 19w09a, 19w12b, 19w13a, 1.14 Pre-Release 2, 1.14 Pre-Release 3, 1.14 Pre-Release 4, 1.14 Pre-Release 5, 1.14.2, 1.14.3 Pre-Release 2, 1.14.3, 1.14.4 Pre-Release 3, 1.14.4 Pre-Release 4, 1.15.2, 20w08a, 20w09a, 20w10a, 20w11a, 20w12a, 20w18a, 20w19a, 1.16 Pre-release 3, 20w46a, 21w10a, 21w11a, 21w13a, 21w15a, 21w17a, 21w18a, 1.17, 1.17.1, 21w37a, 21w40a, 1.18 Pre-release 1, 1.18, 1.18.1, 22w05a, 1.18.2, 22w13a, 22w18a, 22w19a, 1.19 Pre-release 2, 1.19 Pre-release 3, 1.19 Pre-release 4, 1.19 Pre-release 5, 1.19, 1.19.1, 1.19.2, 1.19.3 Pre-release 2, 1.19.3, 23w03a, 23w04a, 1.19.4, 1.20 Pre-release 1, 1.20.1, 23w33a (includes snapshots)

ORIGINAL DESCRIPTION

The bug

When another player is newly rendered on a server (e.g. joining in the server, getting in your render distance (MC-130411

Resolved

), or respawning (

MC-237037

Resolved

)), they will look south until they move their head. This can look pretty strange if the body of the player is facing, for example, north.

Steps to Reproduce

Get two players and label them, "Player A" and "Player B".

- Let player A enter.
- Let player B enter.
- Let player A face north.
- Let player B move far away (beyond render distance) and back.

Observed Behavior

"Player B" sees "Player A" now facing south when actually "Player A" is still facing north, therefore indicating that when newly rendered by the client, all players appear to be facing south until they adjust their head rotation.

Expected Behavior

The expected behavior would be that when players get rendered, all players would appear to be facing the correct direction.

Code Analysis

The following is based on a decompiled version of Minecraft 1.10 using MCP 9.30.

It looks like the method `net.minecraft.entity.Entity.setPositionAndRotation(double, double, double, float, float)` which is called when a player is spawned client-side only sets the body rotation. This could be fixed as well by sending a `net.minecraft.network.play.server.SPacketEntityHeadLook.SPacketEntityHeadLook(Entity, byte)` packet after the spawn packet.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-109346>